Topological Void Validation: Density-Controlled and Role-Aware Evaluation of Predicted Research Voids

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Abstract. Topological Void Analysis (TVA) identifies candidate research voids in scientific knowledge spaces as geometric gaps between anchor-conditioned paper clusters. A natural validation question is whether future papers fill these predicted voids. We show that raw fill rate—whether any future paper appears near a void's midpoint—fails as a validation target for three compounding reasons. First, it is confounded by local corpus density: a hot-zone baseline that samples high-density anchor-adjacent pairs outperforms TVA on raw fill (36% vs. 28%), but this apparent advantage disappears under a density-matched baseline (25%; TVA/B2 lift = 1.12×, 95% CI $[0.85, 1.48]$, not significant). Second, it is limited by anchor observability: only 5–7% of future papers are anchor-eligible per split, so many unfilled voids should be treated as right-censored rather than invalid. Third, even the geometric threshold itself is density-confounded: a fixed threshold of 0.82 corresponds to a 35.7% average null occupancy rate, ranging from 1% to 99% across anchor-density cells. Under a null-calibrated threshold ($\tau_{\text{fill}} \approx 0.841$), fill rates collapse from ∼27% to ∼0.7%, confirming that most raw fills are null-zone noise. Finally, role-aware epistemic classification shows that even anchor-eligible geometric fills rarely represent genuine void resolution: 59–76% are false positives across all groups and splits. These findings motivate a three-layer validation framework—calibrated geometric closure, anchor eligibility, and epistemic role—and establish that raw fill rate measures future local activity near predicted voids but does not validate epistemic closure.

1 Introduction

Classical information retrieval asks: given a query, which existing documents are relevant? Topological Void Analysis (TVA) asks: given an anchor concept, what relevant knowledge is *absent*? TVA identifies candidate research voids as geometric gaps in an embedding-induced knowledge space—regions between anchor-conditioned paper clusters where expected connections, evidence, or conceptual bridges are missing.

Validating predicted voids requires asking: do they correspond to real epistemic gaps, or to geometric artifacts? The natural operationalization is temporal fill rate: does any future paper appear near the void's predicted midpoint? This measure appears transparent, but we show it is confounded at every layer.

Contributions:

1. Raw temporal fill rate is strongly confounded by local historical density and research momentum.

2. A density-matched baseline shows that the apparent TVA disadvantage is not statistically significant (TVA/B2 lift 1.12×, CI $[0.85, 1.48]$).

3. Fixed cosine thresholds are not portable: 0.82 has 35.7% average null FPR; calibrated thresholds collapse raw fill to ∼0.7%.

4. Anchor eligibility exposes an observability limit: only ~6% of future papers are anchor-eligible; unfilled voids are right-censored, not invalid.

5. Role-aware classification confirms that geometric fill substantially overestimates epistemic resolution (59–76% FP across all groups).

Scope. This paper concerns static retrospective validation against a pre-committed future corpus window. Dynamic event-driven extensions are outside scope.

2 Background

2.1 Topological Void Analysis

TVA operates on a corpus embedded with BGE-M3 ($d = 1024$, cosine similarity). For anchor q and source papers A, B, a candidate void satisfies four conditions: (C1) domain cohesion; (C2) calibrated marginality; (C3) sparse lexical bridge; (C4) midpoint vacancy. The midpoint is:

$$m(A, B) = \frac{\mathbf{a} + \mathbf{b}}{\|\mathbf{a} + \mathbf{b}\|} \in \mathbb{S}^{1023}.$$

2.2 Temporal Validation in LBD

Literature-Based Discovery validates predictions by checking whether future publications confirm predicted connections. Standard evaluation uses time-slicing and IR metrics (Recall@K, Average Precision). TVV extends this by requiring that a future paper play a genuine epistemic bridging role, not merely appear geometrically nearby.

2.3 Research Momentum and Dense-Region Confounding

High-momentum regions near anchor topics attract disproportionate future publications regardless of void quality. Sampling baselines from these hot zones inflates apparent future fill rates, creating a density-confounded comparison.

3 Experimental Setup

3.1 Corpus and Temporal Splits

arXiv metadata (cs.OS/AR/PL/SE/DC/NI/CR/AI/LG), three rolling splits:

Table 1: Temporal splits.

Split	T	Val window	Train	Val
t3	2014	2015–2019	~29k	~101k
t4	2016	2017–2021	~35k	~171k
t5	2018	2019–2023	~51k	~193k

3.2 TVA Void Generation

10 anchor concepts × top-30 MMR voids = 300 candidate voids per split.

3.3 Baselines

B1 (hot-zone): 20 random pairs from top-300 anchor C1 pool per anchor. Biased toward high-density, high-momentum regions.

B2 (density-matched): For each TVA void, select the B1-style pair with midpoint local density $\rho(m) = \frac{1}{k}\sum \mathrm{sim}(m, \mathrm{kNN}_i(m))$ ($k = 20$) closest to the TVA void's midpoint density.

3.4 Three-Layer Fill Predicate

Anchor-eligible val papers:

$$\mathrm{Val}_q = \{P \in \mathrm{Val}_t : \mathrm{sim}(P, q) \geq \tau_q\}, \quad \tau_q = \max(Q_{80}^{\mathrm{val}}, Q_{90}^{\mathrm{train}})$$

Geometric filler candidate: $P_V^* = \arg\max_{P \in \mathrm{Val}_q} \mathrm{sim}(P, m_V)$

Geometric closure:

$$G(V) = \mathbf{1}[\mathrm{sim}(P_V^*, m_V) \geq \tau_{\mathrm{fill}}(q, \rho(V), t)]$$

Validated epistemic fill:

$$\mathrm{Fill}(V) = G(V) \wedge R(P_V^*, A, B, q)$$

where $R = 1$ iff role $\in \{\text{TRUE-FILL}, \text{PARTIAL-FILL}\}$.

Calibrated threshold:

$$\tau_{\mathrm{fill}}(q, \rho, t) = Q_{1-\alpha}\big(\{\max_{P \in \mathrm{Val}_q} \mathrm{sim}(P, m_0) : m_0 \in \mathrm{Null}(q, \rho, t)\}\big), \quad \alpha$$

4 Results

4.1 Finding 1: Raw Fill Rate Is Density-Confounded

Table 2: Raw fill rates under legacy $\tau = 0.82$. CI from 1000 bootstrap resamples (t5).

Split	TVA	B1 (hot)	B2 (density)	TVA/B2
t3	31%	32%	30%	1.03×
t4	26%	32%	23%	1.12×
t5	27%	36%	24%	1.12×

B1 consistently outperforms TVA by up to 1.30×. After density matching (B2), TVA achieves 1.03–1.12× lift, but the t5 CI includes 1.0—not significant.

4.1.1 Fixed Thresholds Are Not Portable

Table 3: Calibrated threshold and null FPR@0.82 by density bucket (t5 mean).

Density bucket	Null p50	τ_{fill} (p95)	FPR@0.82
Low	0.787–0.811	0.790–0.859	1–18%
Mid	0.802–0.832	0.818–0.861	5–80%
High	0.810–0.854	0.841–0.881	28–99%
Global mean	–	**0.841**	**35.7%**

Under calibrated τ_{fill}, fill rates collapse: TVA: 27.3%→**0.7%**; B2: 24.3%→**0.7%**; lift: 1.00×.

The 0.7% < 5% result is correct: TVA C1–C4 conditions select sparser bridge regions than random density-matched pairs, so fewer reach the calibrated threshold. Most raw fills at $\tau = 0.82$ are null-zone occupancy, not significant geometric closure.

4.2 Finding 2: Anchor Eligibility Reveals Observability Limits

Only 5–8% of val papers are anchor-eligible per split. Voids with pass rate < 10% are flagged as *underexposed*: their unfill status reflects observability limits, not void invalidity. ebpf_obs

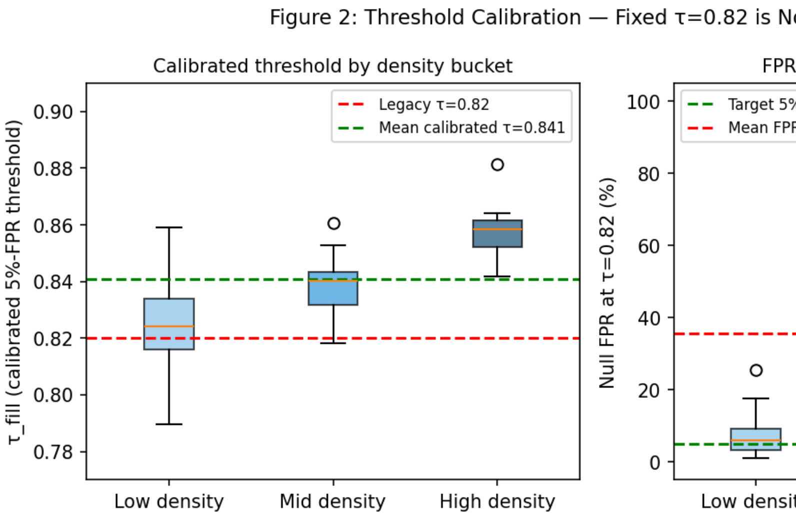

Figure 1: Calibrated threshold by density bucket (left) and null FPR at 0.82 (right). The fixed threshold 0.82 has 35.7% average null FPR, reaching 99% in high-density anchor cells. Calibrated $\tau_{\text{fill}} \approx 0.841$ is stable across all splits.

Table 4: Anchor exposure (t5 sample). Mean sim and hybrid gate pass rate.

Anchor	Mean sim	τ_q	Pass rate	TVA fill
sched_opt	0.520	0.582	5.8%	40%
ebpf_obs	0.495	0.547	6.5%	0%
virt_hyp	0.498	0.555	7.7%	57%
Mean	**0.503**	**0.563**	**6.2%**	–

(6.5% pass, 0% fill) illustrates that non-zero exposure does not guarantee eligible papers overlap predicted void neighborhoods.

4.3 Finding 3: Geometric Fill Rarely Equals Epistemic Fill

59–76% of gated raw fills are false positives across all groups and splits. Near-miss controls show the highest FP rate (84%) and lowest score, confirming that geometric proximity carries some signal but is insufficient. TVA and B1 are statistically indistinguishable; B2 performs marginally better.

4.4 Case Studies

Three representative cases from t4/t5:

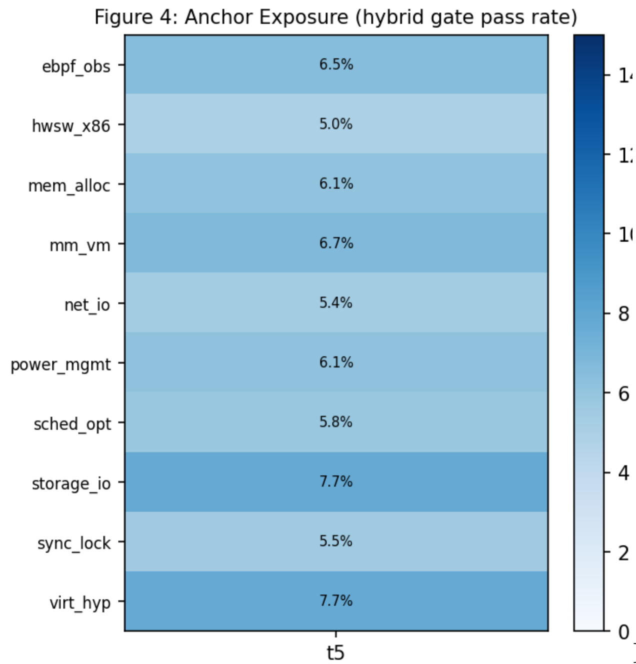

Figure 2: Anchor eligibility pass rate (%) by anchor and split (hybrid gate). Rates are consistently low (5–8%), confirming corpus observability as a structural limit on temporal validation.

Table 5: Epistemic role decomposition (t3/t4/t5 combined, 500 sampled cases). Void res. = TRUE-FILL + PARTIAL-FILL; Boundary = IE + SE + SN; FP = FALSE-POSITIVE.

Source	n	Void res.	Boundary	FP	Score s
TVA (gated)	98	0%	33%	69%	0.068
B1 (gated)	73	0%	33%	70%	0.062
B2 density	107	3%	36%	62%	0.104
Near-miss	219	2%	13%	84%	0.043

Case 1 – Partial Fill (arXiv:1912.06243, 2019, 18 citations): Vehicular cloud multi-task offloading. $\mathrm{sim}(P, m) = 0.835$. Contains 10+ task-offloading refs (B-side) but only 1 marginal virtualization ref (A-side). Role: PARTIAL-FILL—one-sided bridging.

Case 2 – Boundary Expansion (arXiv:2106.01441, 2021, 7 citations): AI planning for heterogeneous system optimization. $\mathrm{sim}(P, m) = 0.855$. Zero kernel-scheduler refs; all 44 refs in heterogeneous HPC. Role:

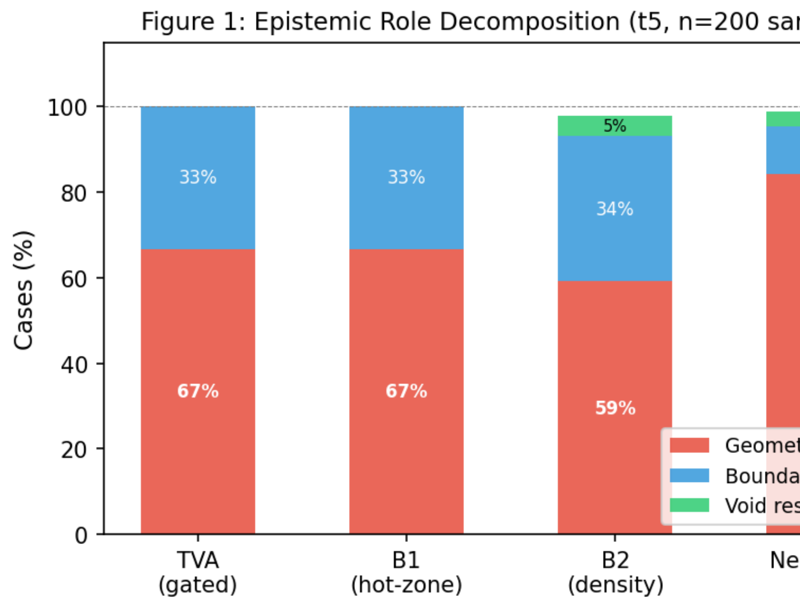

Figure 3: Epistemic role decomposition by source group (t5, 200 sampled cases). Most raw fills are geometric false positives regardless of method. Near-miss controls show the highest FP rate, confirming that geometric proximity is a weak but non-zero signal.

INCREMENTAL-EXTENSION—pure boundary expansion along B-direction.

Case 3 – False Positive (arXiv:1710.11590, 2017, 6 citations): Energy-aware virtual network embedding. $\mathrm{sim}(P, m) = 0.875$ (highest). Zero storage I/O or NVMe refs. Anchor is storage I/O—completely unrelated. Role: FALSE-POSITIVE—dense-region artifact.

5 Discussion

5.1 Raw Fill Fails for Three Reasons

Raw fill conflates void quality with sociotechnical realization rate: (1) hot-zone baselines are biased by local momentum; (2) fixed thresholds are not portable across anchor-density regimes; (3) anchor-eligible future papers are sparse (∼6%), right-censoring many unfilled voids.

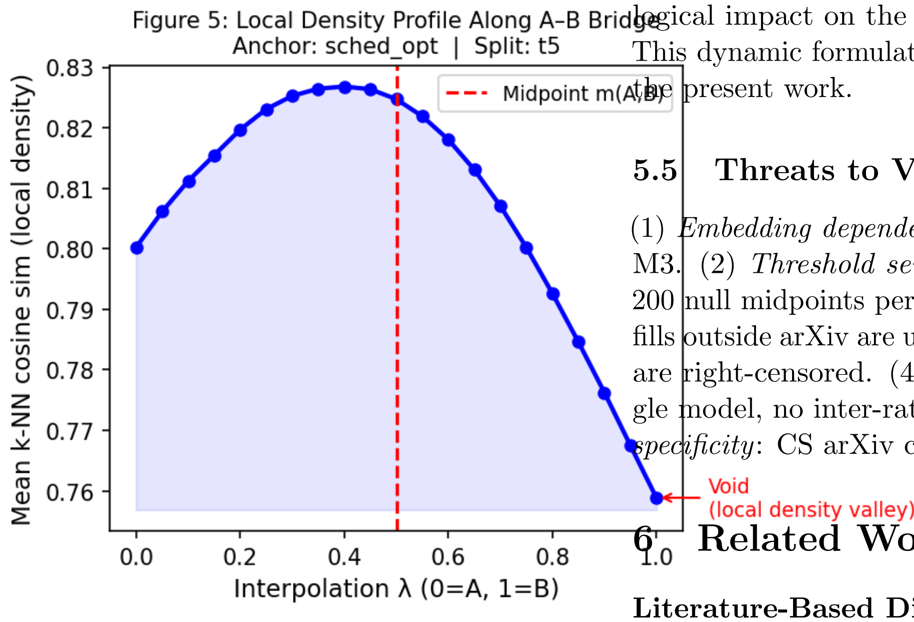

Figure 4: Local density profile along the A–B bridge for the highest-MMR void in anchor sched_opt (t5). The density valley at $\lambda \approx 0.5$ marks the void midpoint region—two dense source communities with a less-occupied bridge.

5.2 From Geometric Occupancy to Role Decomposition

Future papers near void midpoints decompose into true fills, boundary expansions, and false positives. The scarcity of true fills reveals that genuine epistemic closure is a much stricter condition than geometric occupancy. *Geometric closure and anchor eligibility are screening conditions; epistemic role is the validation condition.*

5.3 Descriptive, Not Evaluative

TVV classifies the type of validation evidence produced by future literature, not the intrinsic merit of papers. Boundary expansion and incremental extension are valuable scientific activities.

5.4 Dynamic Extension (Future Work)

These results suggest a natural extension: instead of asking whether a predicted void is eventually filled within a fixed window, one can classify each newly arriving paper by its pre-insertion topological impact on the existing knowledge space. This dynamic formulation is outside the scope of the present work.

5.5 Threats to Validity

(1) *Embedding dependence*: all metrics use BGE-M3. (2) *Threshold sensitivity*: calibration uses 200 null midpoints per cell. (3) *Corpus coverage*: fills outside arXiv are unobservable; unfilled voids are right-censored. (4) *LLM classification*: single model, no inter-rater reliability. (5) *Domain specificity*: CS arXiv categories only.

6 Related Work

Literature-Based Discovery. Swanson's ABC model and subsequent LBD work validate predictions by checking future publication appearance. TVV extends this tradition by requiring epistemic bridging role, not merely geometric proximity.

Novelty and First-Story Detection. Topic Detection and Tracking introduced first-story detection; novelty detection in IR measures whether a document adds new information relative to a query set. Our first-filler intuition shares the event-ordering structure but operates on knowledge structure rather than event reporting.

Structural Holes and Scientometrics. Burt's structural hole theory shows brokers spanning gaps accumulate informational advantage. Our density-matched baseline directly controls for hot-zone momentum.

Dense-Region Bias in Embeddings. Anisotropy in high-dimensional text embeddings compresses cosine similarities into a narrow range. We show this anisotropy makes fixed thresholds non-portable in downstream void validation.

Scientific Claim Verification. Epistemic classification of scientific contributions—argumentative zoning, overclaiming detection—provides precedents for our role-aware rubric.

7 Conclusion

Temporal validation of predicted research voids requires separating geometric occupancy, density bias, anchor observability, and epistemic role. Under density matching, TVA's apparent disadvantage disappears but is not statistically significant. Under null-calibrated thresholds, fill rates collapse to $\sim$0.7%—most raw fills were null-zone noise. Role-aware classification confirms that genuine epistemic bridging is rare even among geometrically close, anchor-eligible future papers.

We propose a three-layer validation framework and leave two open questions for future work: (1) scalable role-aware validation without prohibitive LLM cost; (2) dynamic event-driven extensions in which each arriving paper updates knowledge-space state.

Acknowledgements. TODO.

References